



Darshan University

A Project Report on

“Blood Bank management System”

Under the subject

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B. Tech, Semester – IV

Computer Science & Engineering Department

Submitted By

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Internal Guide

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**Computer Science & Engineering
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DECLARATION

We hereby declare that the SRS, submitted along with the **Software Engineering (2301CS405)** for entitled “**Blood Bank Management System**” submitted in partial fulfilment for the Semester-5 of **Bachelor Technology (B. Tech)** in **Computer Science and Engineering (CSE)** Department to Darshan University, Rajkot, is a record of the work carried out at **Darshan University, Rajkot** under the supervision of R. B. Gondaliya and that no part of any of report has been directly copied from any students’ reports, without providing due reference.

(Tushar Bavaliya)

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Student’s Signature

Date: _____



**Computer Science & Engineering
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CERTIFICATE

This is to certify that the SRS on “**Blood Bank Management System**”
has been satisfactorily prepared by **Tushar Bavaliya (24010101015)** under
my guidance in the fulfillment of the course **Software Engineering
(2301CS405)** work during the academic year 2025-2026.

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Thus, in conclusion to the above said, I once again thank the faculties and members of **Darshan University** for their valuable support in completion of the project.

Thanking You

Tushar Bavaliya

ABSTRACT

The Blood Bank Management System is designed to computerize and streamline all daily operations of a blood bank. It helps transform the traditional manual process into an efficient digital system. The system includes important features such as user login, quick blood search, and admin access to monitor all activities. The administrator or staff can add, view, update, and delete donor details, blood units, and hospital requests. Donors can log in to view their donation history, eligibility, and upcoming appointments, as well as request donation slots or inquire about needed blood types. The admin can also generate various reports, such as donor reports, blood stock reports, request reports, and hospital usage reports.

Main purpose of this system is to reduce human efforts as much as possible.

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1 Introduction

1.1 Product perspective

This project focuses on converting the manual blood bank process into an internet-based application, allowing users to easily access information such as blood availability and their donation details. It is a multi-user system capable of handling all core operations of a blood bank, including donor management, blood collection, screening, inventory control, and request processing. The system efficiently supports all essential functions required for a small to medium blood bank.

1.2 Product features

1.2.1 There are three different users who will be using this product:

- Administrator who will manage Blood Bank.
- Donor who will be Donate Blood.
- Acceptor who will request for Blood.

1.2.2 The features that are required for the Administrator are:

- Can approve or reject donor eligibility based on medical test results.
- Can manage blood testing records (HIV, Hepatitis, Blood grouping, etc.).
- Can schedule blood donation camps and manage camp details.
- Can generate expiry alerts for blood units nearing expiration.
- Can track blood transfer history between blood banks or hospitals.
- Can manage staff accounts and assign system roles.
- Can maintain emergency blood request records.
- Can view daily, weekly, and monthly blood usage reports.
- Can backup and restore blood bank data.
- Can ensure compliance with blood bank rules and regulations.
- Can lock or deactivate inactive donor or hospital accounts.
- Can maintain audit logs for all blood bank transactions.

1.2.3 The features that are required for the Donor are:

- Register themselves in the blood bank system.
- Update personal and contact information when required.
- View eligibility status for blood donation.
- Search for nearby blood banks or donation camps.
- Book or cancel blood donation appointments.
- View donation history and last donation date.
- Receive notifications for upcoming donation camps and reminders.
- Check blood test results after donation.
- Track reward points or certificates (if applicable).
- Communicate with the blood bank for queries or support.

1.2.4 The features that are required for the Acceptor are:

- Register themselves in the blood bank system.
- Search for available blood based on blood group and quantity.
- Request blood units for personal or patient use.
- View the status of blood requests (approved / pending / rejected).
- Maintain patient and hospital details.
- Track issued blood units received from the blood bank.
- View request history and transfusion details.
- Receive notifications for request approval or availability.

1.3 Functional Requirement

1.3.1 Librarian

- Manage users
- Register Staff and Donor
- Manage roles
- View donor list
- View Acceptor List
- View full Blood inventory
- Generate reports
- Monitor stock alerts
- Manage hospitals
- View activity logs
- Configure system settings
- Manage notifications
- Oversee collections
- Oversee requests
- Backup database
- Manage campaigns
- Approve inventory changes
- View system dashboard
- Manage security settings

1.3.2 Donor

- register account
- Edit profile
- View donation history
- Schedule donation
- Get reminders
- View blood group
- Cancel appointments
- Receive notifications
- Check eligibility
- Digital donor ID
- Download certificate
- Join donation drives
- Receive health tips
- Find blood centres

- Give feedback
- Track health vitals
- Get emergency alerts

1.3.3 Acceptor

- Register Account
- Edit profile
- View accept history
- Login
- View blood group
- Check blood availability
- Search blood
- Request blood
- Select blood bank
- Upload prescription
- Submit medical details
- Track request status
- Receive notification
- Cancel request
- View request history
- Update profile
- Contact blood bank
- View blood bank details
- Get emergency alerts

1.4 Non-Functional Requirement

1.4.1 Usability:

- The UI should be simple enough for everyone to understand and get the relevant information without any special training. Different languages can be provided based on the requirements.

1.4.2 Accuracy:

- The data stored about the books and the fines calculated should be correct, consistent, and reliable.

1.4.3 Availability:

- The System should be available for the duration when the library operates and must be recovered within an hour or less if it fails. The system should respond to the requests within two seconds or less.

1.4.4 Maintainability:

- The software should be easily maintainable and adding new features and making changes to the software must be as simple as possible. In addition to this, the software must also be portable.

2 Design and Implementation Constraints

2.1 Use case diagram

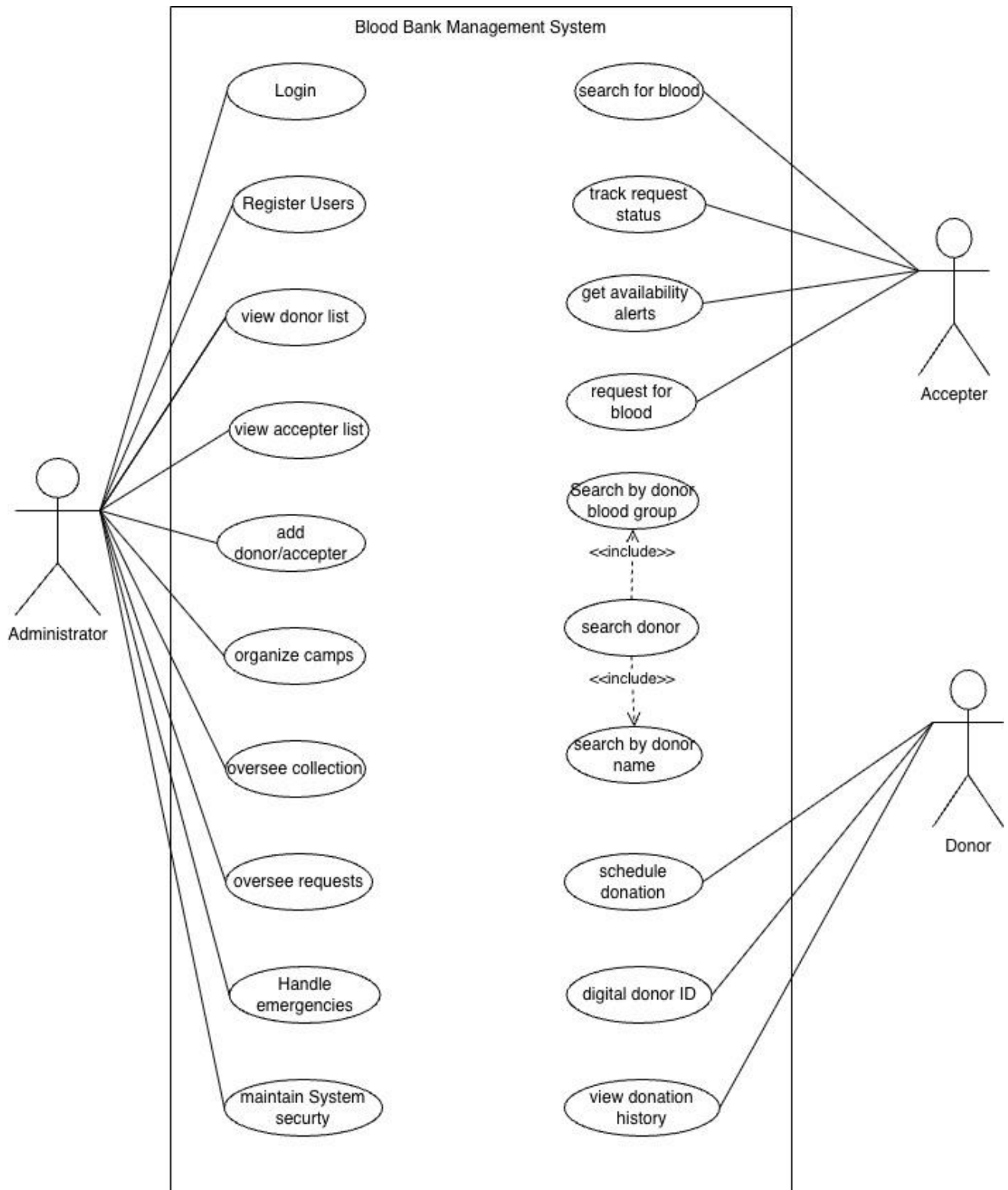


Figure 2.1-1 Use case diagram for Blood Bank management system

2.2 Activity diagram and Swimlane diagram

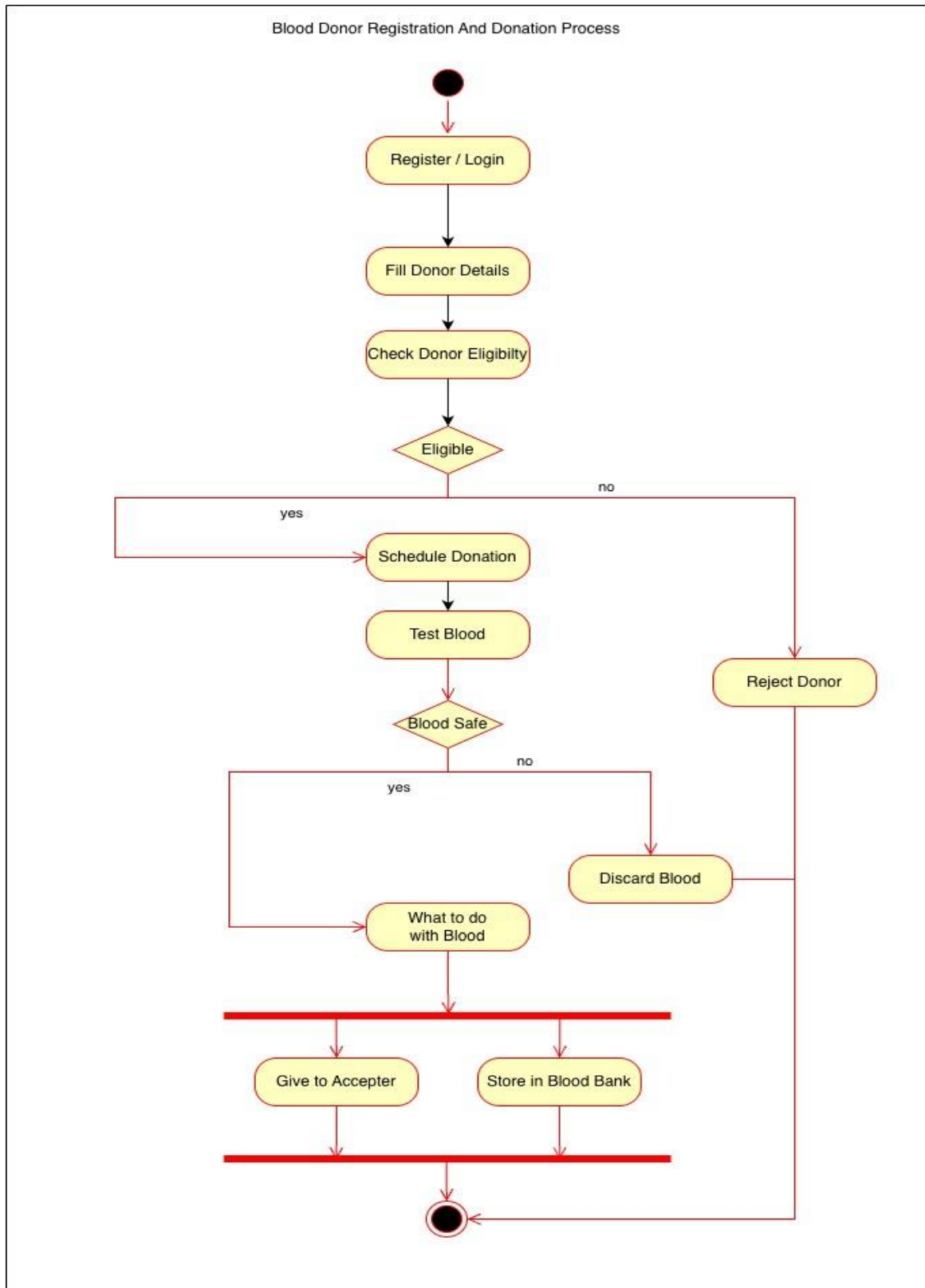


Figure 2.2-1 Activity diagram for Blood Donor Registration And Donation Process

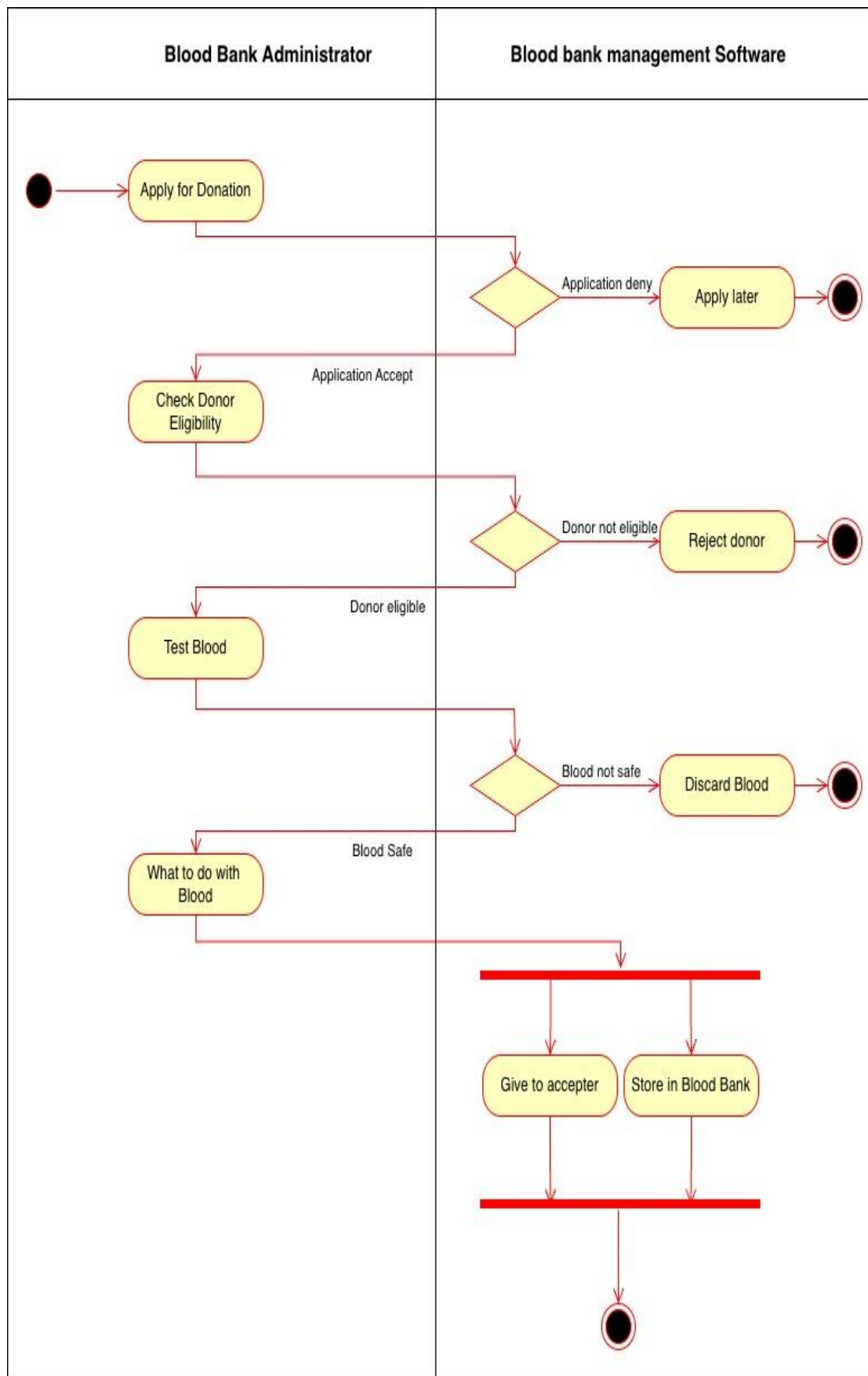


Figure 2.2-2 Swimlane diagram for Blood Donate

2.3 Sequence diagram

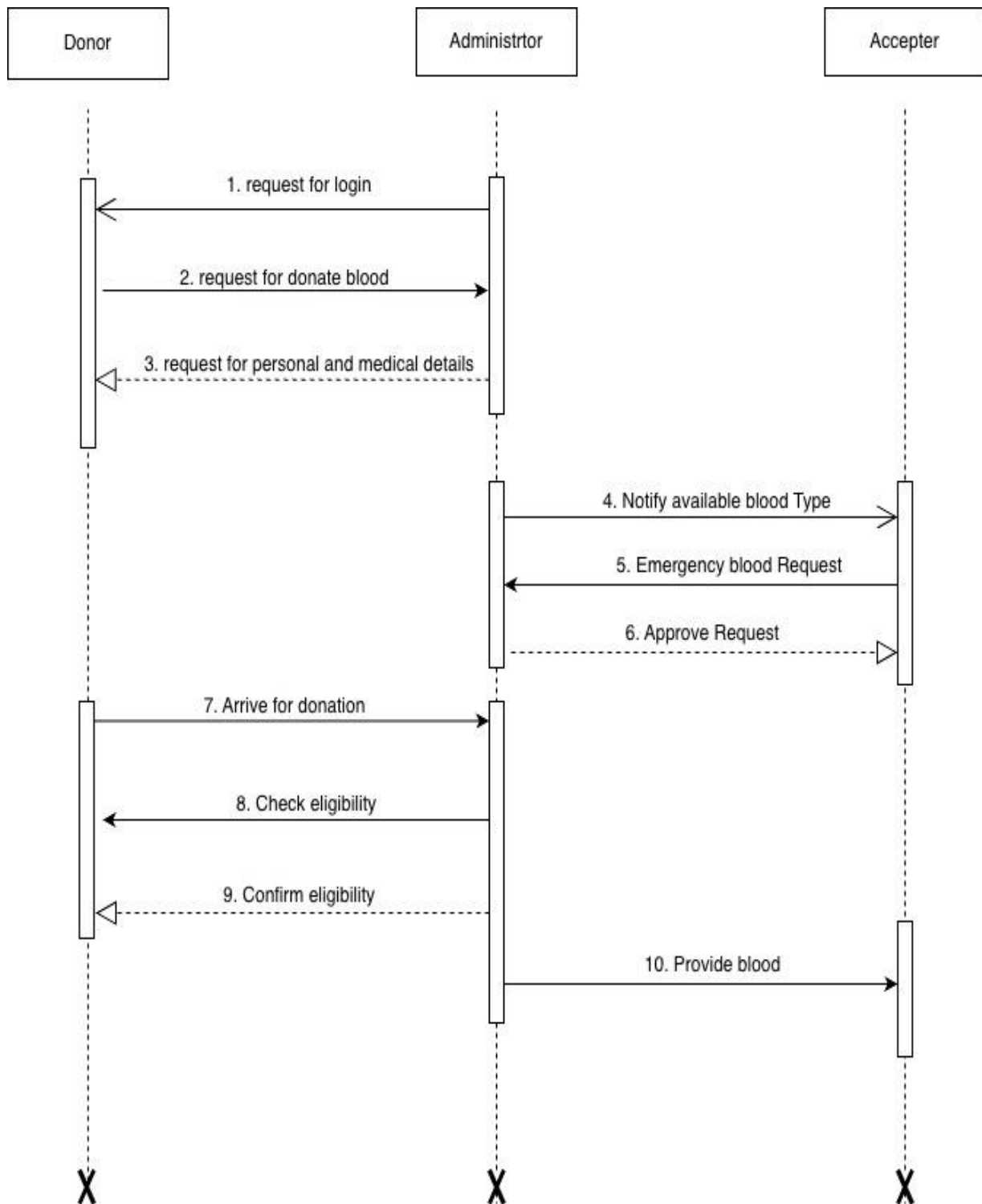


Figure 2.3-1 Sequence diagram for Blood donation

2.4 State diagram

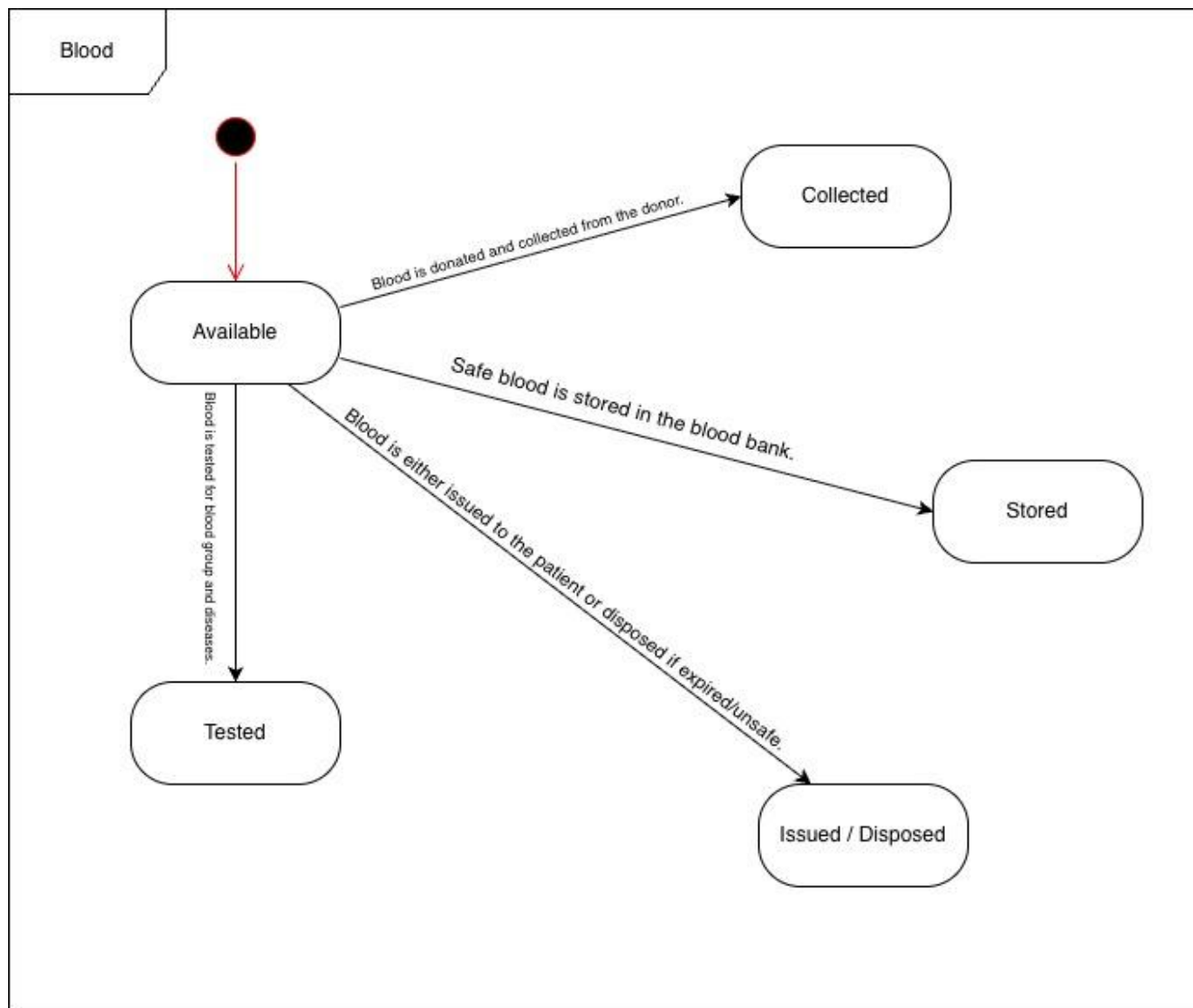


Figure 2.4-1 State diagram of Blood

2.5 Class diagram

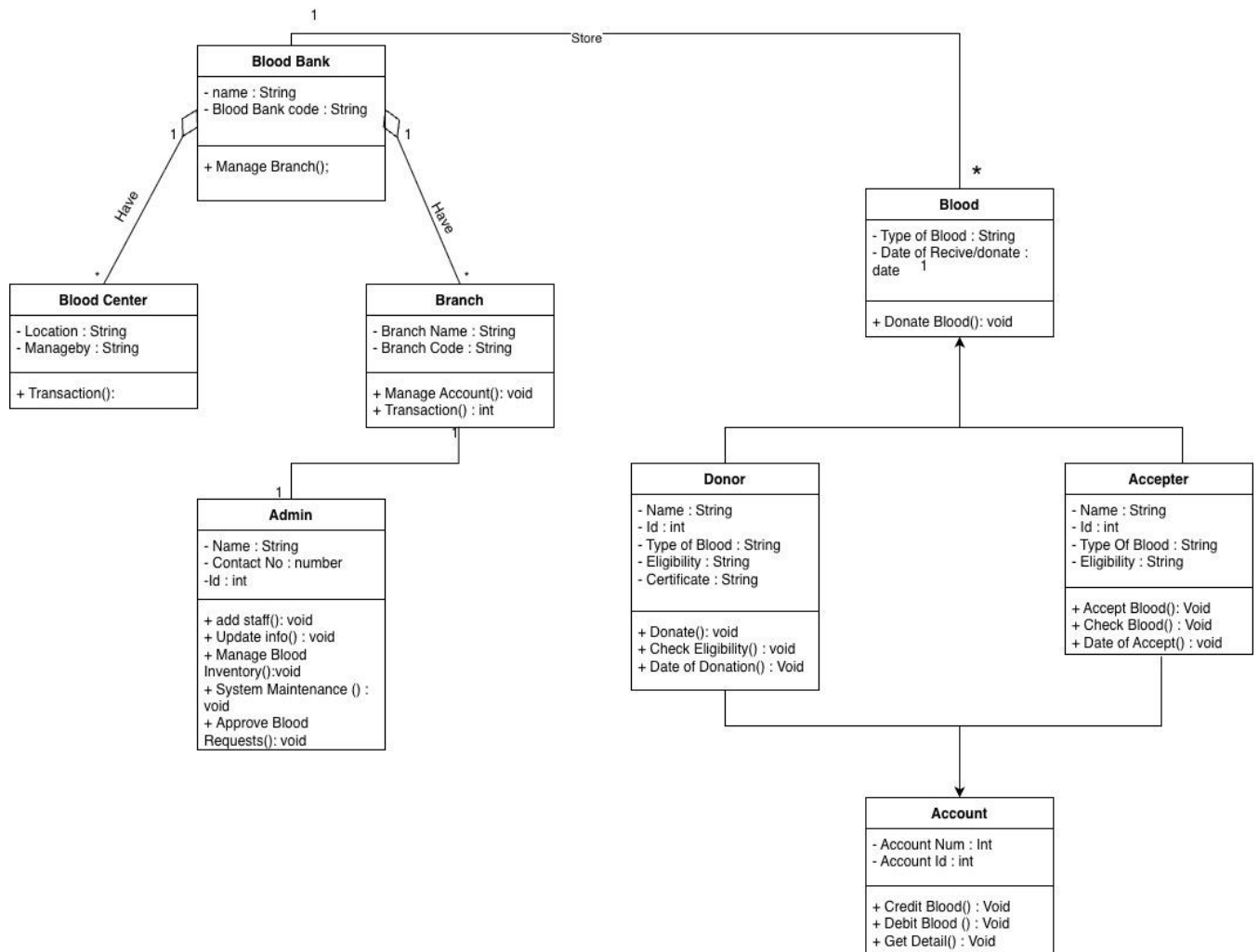


Figure 2.5-1 Class diagram for Blood Bankmanagement system

2.6 Data flow diagram

2.6.1 Context diagram (level-0)

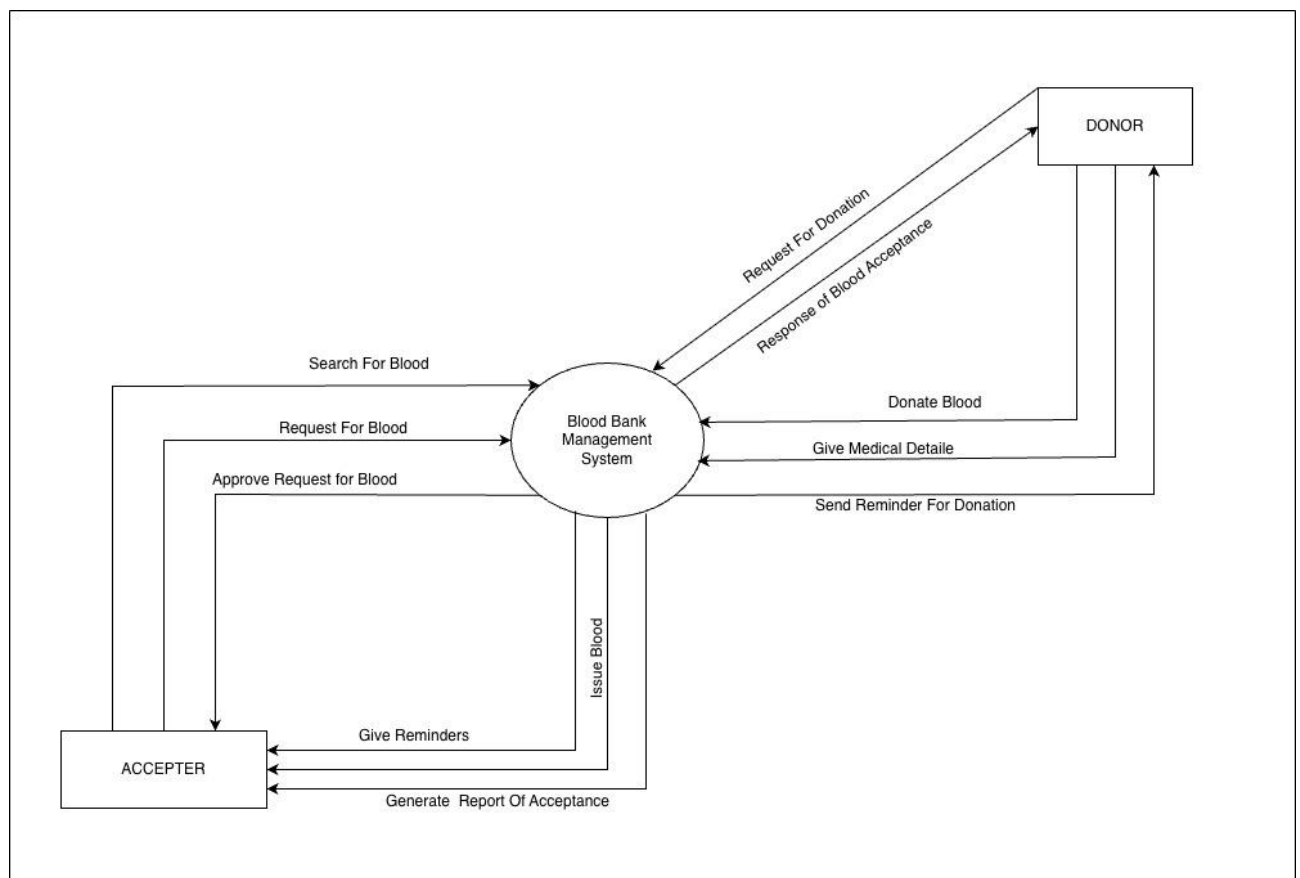


Figure 2.6-1 Context diagram for Blood Bank management system

2.6.2 DFD Level-1

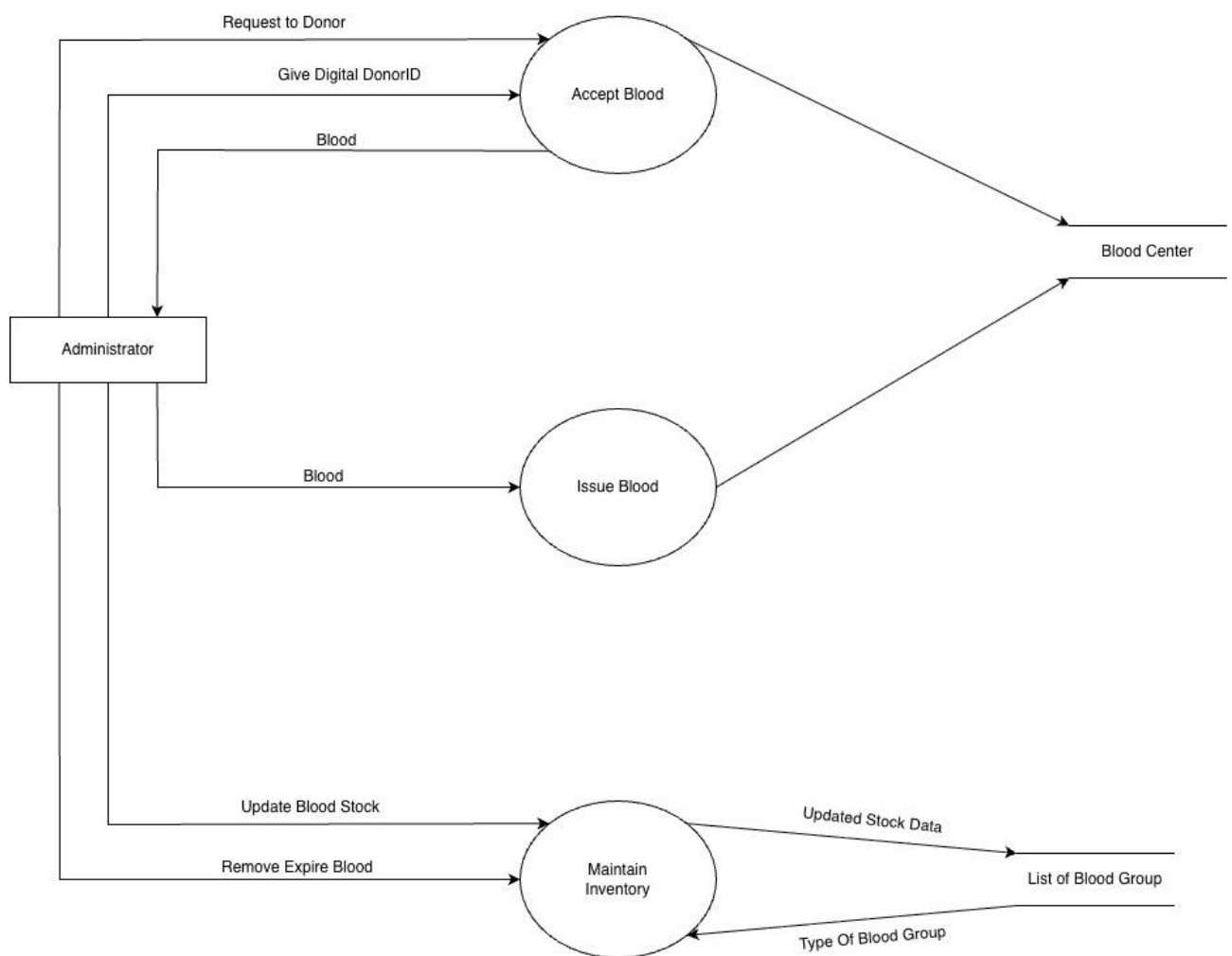


Figure 2.6-2 DFD level-1 for Blood management system

2.6.3 DFD Level-2

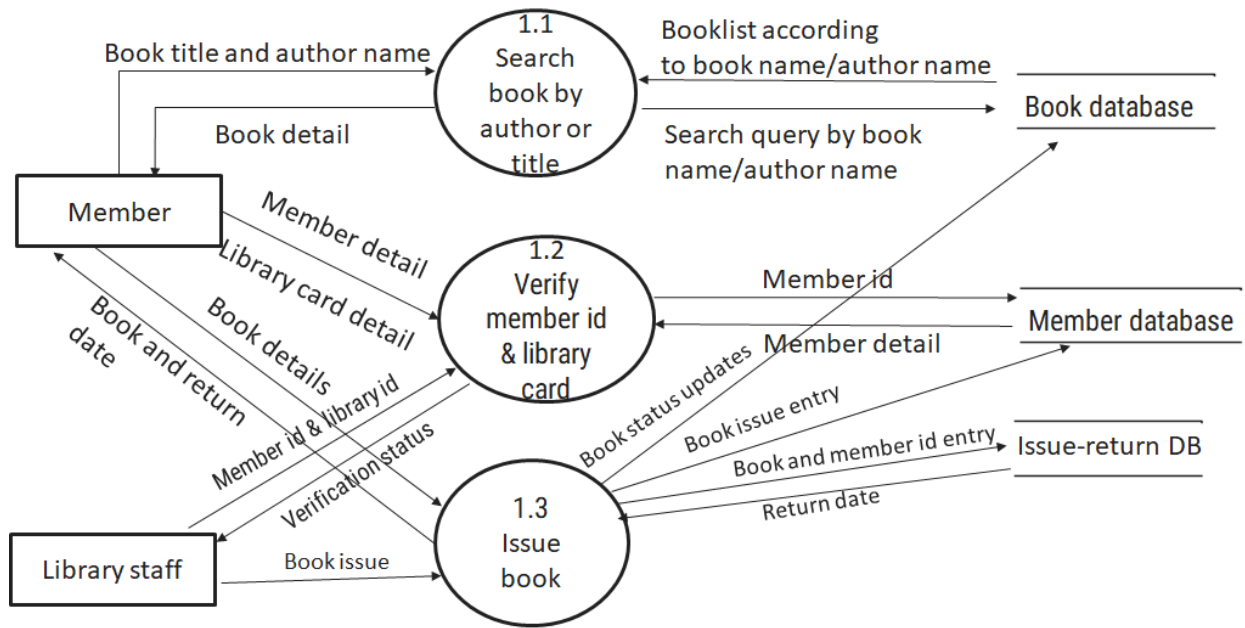
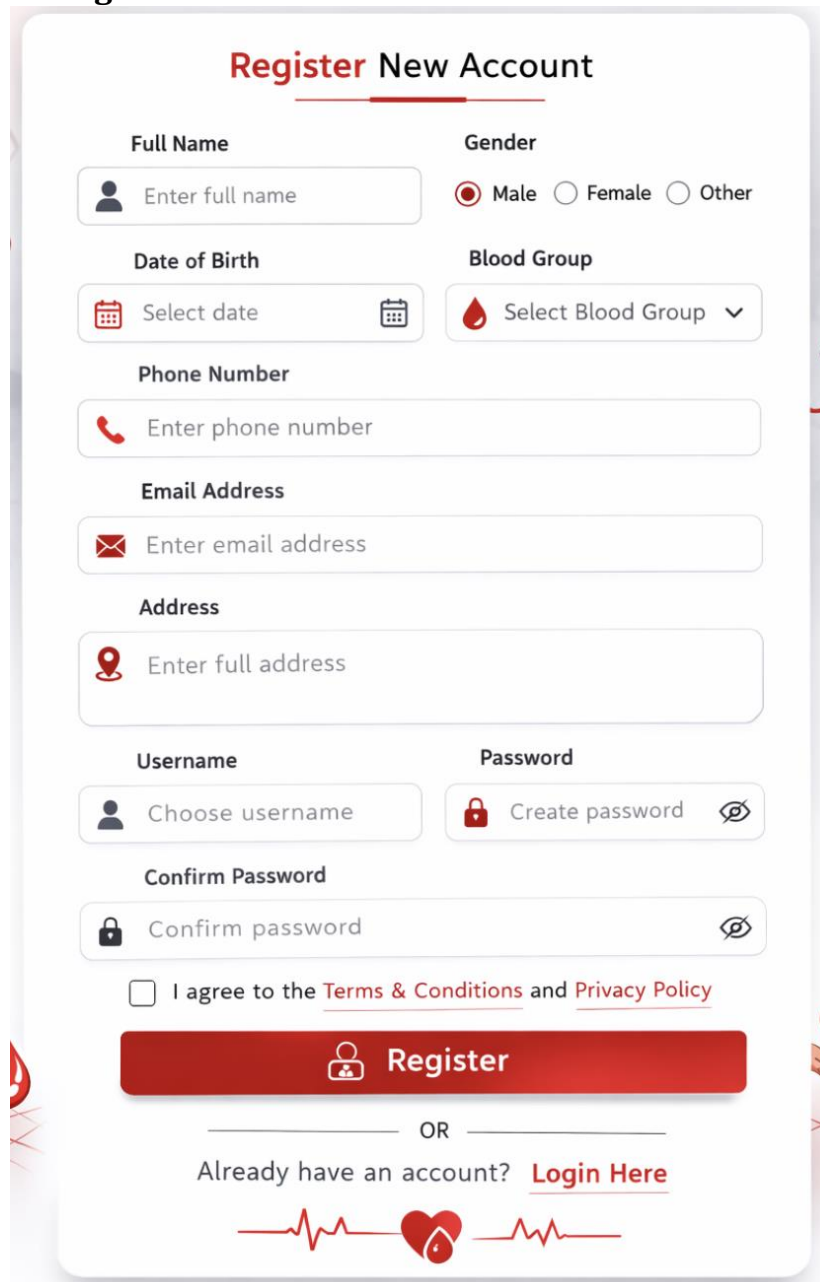


Figure 2.6-3 DFD level-2 for Issue book

3 External interface requirement (Screens)

3.1 Screen-1: Registration Form



The screenshot shows a 'Register New Account' form with the following fields and options:

- Full Name:** Text input field with a person icon.
- Gender:** Radio buttons for Male (selected), Female, and Other.
- Date of Birth:** Date picker with a calendar icon.
- Blood Group:** Dropdown menu with a blood drop icon.
- Phone Number:** Text input field with a phone icon.
- Email Address:** Text input field with an envelope icon.
- Address:** Text input field with a location pin icon.
- Username:** Text input field with a person icon.
- Password:** Text input field with a lock icon and a toggle for visibility.
- Confirm Password:** Text input field with a lock icon and a toggle for visibility.
- Agreement:** A checkbox followed by the text 'I agree to the [Terms & Conditions](#) and [Privacy Policy](#)'.
- Register Button:** A red button with a person icon and the text 'Register'.
- OR:** A horizontal line with the text 'OR' in the center.
- Login Link:** The text 'Already have an account? [Login Here](#)'.
- Decorative Element:** A red heart icon with a blood drop inside, flanked by red ECG lines.

Figure 3.1-1 Screen-1: Registration Form

Purpose: This form will allow the target end-users to register in the system. To register , the following information will be encoded in the system.

Table 3.1-1 Screen element of Registration form

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Full Name	Textbox	M	1	Full Name field should be editable and accept alphabetic characters.

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2	Gender	Radio Button	M	1	User must select one option (Male / Female / Other).
3	Date of Birth	Date Picker	M	1	User should select date using calendar input.
4	Blood Group	Dropdown	M	1	User must select blood group (A+, A-, B+, etc.).
5	Phone Number	Textbox	M	1	Should accept valid numeric phone number (10 digits).
1	Username	Textbox	M	1	Username field should be editable and accept the Username.
2	Email	Textbox	M	1	Email field should be editable and accept the email with proper format.
3	Password	Password	M	1	Password field should be editable and accept the password and display as star or dot.
4	Retype password	Password	M	1	Retype Password field should be editable and accept the password and display as star or dot.
5	I agree to the terms	Checkbox	M	1	Checkbox for accepting terms and condition.
6	Already have an Account	Link	-----	-----	Link for navigate to login page for registered user.
7	Register	Button	-----	-----	Register is a button for store the entered data into database.

3.2 Screen-2: Login Form

Login to Your Account

Username or Email

Password

☐ Remember Me [Forgot Password?](#)

Login

Don't have an account?
Register Here

Figure 3.2-1 Screen-2: Login Form

Purpose: This form will be used by the system's users to access records and features of the system. The users will input the correct combination of their username and password to be able to login to the system.

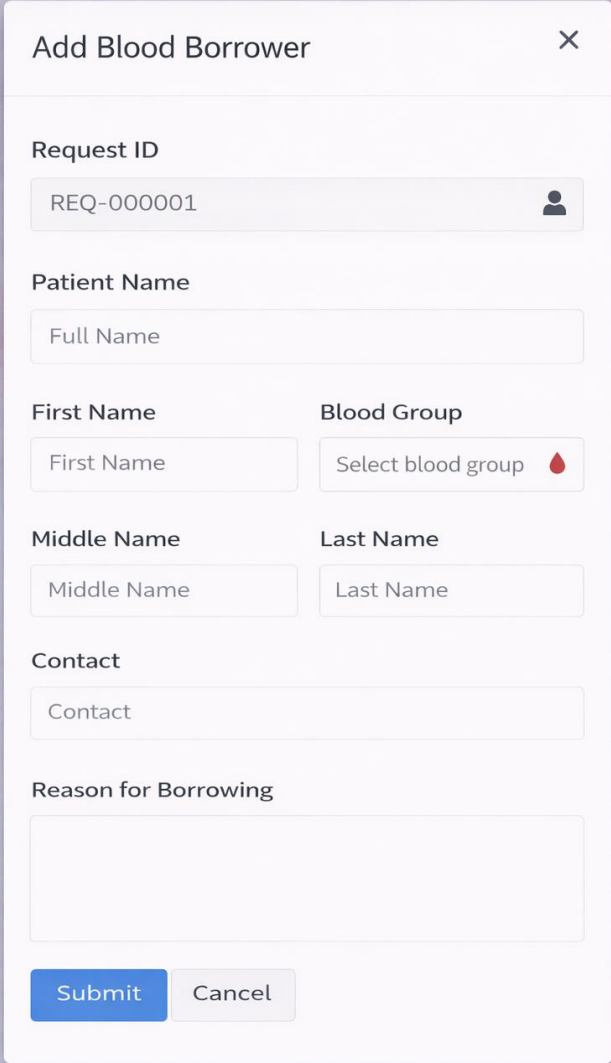
Table 3.2-1 Screen element of Login form

Sr.	Screen Element	Input Type	O/M	1/N	Description
-----	----------------	------------	-----	-----	-------------

SRS – Blood Bank Management System

1	Username	Textbox	M	1	Username field should be editable and accept the Username.
2	Password	Password	M	1	Password field should be editable and accept the password and display as star or dot.
3	Remember Me	Checkbox	M	1	Saving login credentials through remember me checkbox
4	forgot password	Link	-----	-----	Link for navigate to Forgot password page for allows users to recover password.
5	Register Here	Link	-----	-----	Link for navigate to membership registration.
6	Log in	Button	-----	-----	Login button navigates to another page even if valid login credentials.

3.3 Screen-3: Add Blood borrower



Add Blood Borrower [X]

Request ID
REQ-000001 [User Icon]

Patient Name
Full Name

First Name **Blood Group**
First Name Select blood group [Blood Drop Icon]

Middle Name **Last Name**
Middle Name Last Name

Contact
Contact

Reason for Borrowing
[Text Area]

Submit **Cancel**

Figure 3.3-1 Screen-3: Add Blood borrower

Purpose: This module will allow the system administrator to add, edit, update or delete borrowers of Blood . The admin can add borrower information and manage it.

Table 3.3-1 Screen element of Request Blood

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Request ID	Textbox	M	1	Request ID field should be auto-generated and editable for reference.
2	Patient Name	Textbox	M	1	Patient Name field should be editable and accept only alphabets.
3	First Name	Textbox	M	1	First Name field should be editable and accept only string as a borrower first name.
4	Middle Name	Textbox	O	1	Middle Name field should be editable and accept only string as a borrower middle name.
5	Last Name	Textbox	M	1	Last Name field should be editable and accept only string as a borrower last name.
6	Contact	Textbox	M	1	Contact field should be editable and accept only 10-digit mobile number.
7	Blood Group	Drop Down	M	1	User must select blood group (A+, A-, B+, etc.).
8	Submit	Button	-----	-----	Submit is a button for store the entered data into database.
9	Cancel	Button	-----	-----	Cancel is a button for close add borrower dialog.

3.4 Screen-4: Blood requests

Blood Requests						View All >
ID	Patient Name	Blood Group	Units	Status	Request Date	
REQ-000182	Ethan Carter	B+	2	Pending	Apr 22, 2024	
REQ-00180	Sarah Foster	A+	2	Approved	Apr 20, 2024	
REQ-00179	Michael Brown	O-	2	Pending	Apr 20, 2024	
REQ-00176	Laura Wilson	AB+	2	Rejected	Apr 18, 2024	
REQ-00175	Daniel Clark	A-	2	Pending	Apr 18, 2024	
Showing 5 of 18				Showing 5 of 18 < >		

Figure 3.41 Screen-3: Blood Requests

Table 3.41 Screen element of Blood Requests

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	ID	Label (Table Column)	M	N	Displays unique request ID for each blood request.
2	Patient Name	Label (Table Column)	M	N	Displays name of the patient requesting blood.
3	Blood Group	Label (Table Column)	M	N	Displays required blood group (e.g., B+, A+, O-).
4	Units	Label/Badge	M	N	Displays number of units required (if included).
5	Status	Label	M	N	Displays Status of Blood request.
6	Request Date	label	M	N	Displays The date of request blood.

3.5 Screen-5: Add Blood borrower

Recent Donors

View All >

Donor Name	Blood Group	Units	Last Donation
Emily Davis	 B+	2	Aprl 23, 2024
James Wilson	 A-	2	Aprl 20, 2024
Olivia Martin	 O+	2	Aprl 20, 2024
Benjamin Lee	 AB-	2	Aprl 18, 2024
Emma Harris	 A-	2	Aprl 14, 2024

Showing 5 of 29

Showing 5 of 29

< | >

figure 3.51 Screen-3: Recent donors

Table 3.51 Screen element of recent Donor

Sr.	Screen Element	Input Type	O/M	1/N	Description
1	Donor Name	Label (Table Column)	M	N	Displays the name of the blood donor.
2	Last Donation	Label (Date)	M	N	Displays the last donation date of the donor.
3	Blood Group	Label (Table Column)	M	N	Displays required blood group (e.g., B+, A+, O-).
4	Units	Label/Badge	M	N	Displays number of units required (if included).
5	View All Button	Button	O	1	Navigates to full donor list page
6	Pagination	Control (Buttons)	O	1	Allows navigation between donor records pages
7	Record Count Info	Label	O	1	Displays current showing records (e.g., "Showing 5 of 29").

4 Data Base design

4.1 List of Tables

- Blood
- Administrator
- Donor
- Acceptor

Table 4.1-1 Table: Blood

Column	Data Type	Null	Keys & Constrains	Default Value & Description
Blood group	Varchar (3)	NN	PK (Auto Increment)	
Blood code	int	NN		
Expiry date	datetime	NN		
Donor id	int	NN		
Issued Date	datetime	NN		

Table 4.1-2 Table: Acceptor

Column	Data Type	Null	Keys & Constrains	Default Value & Description
Acceptor	int	NN	PK (Auto Increment)	
Acceptor name	varchar (100)	NN		
Blood Group	varchar (3)	NN	FK	Reference of Blood Table
AcceptFromDate	Date Time	AN		
AcceptToDate	DateTime	AN		
Issued date	datetime	NN	FK	Reference of Blood Table
Contact	number (10,0)	AN		

Table 4.1-3 Table: Donor

Column	Data Type	Null	Keys & Constrains	Default Value & Description
DonorId	int	NN	PK (Auto Increment)	
Donor Name	varchar(100)	NN		
BloodGroup	varchar (3)	AN	FK	Reference of Blood Table
Age	int	AN		
Contact	number (10,0)	AN		

Table 4.1-4 Table: administrator

Column	Data Type	Null	Keys & Constrains	Default Value & Description
AdministratorID	int	NN	PK (Auto Increment)	
AdminName	varchar (100)	NN		
Gender	varchar (100)	AN		
DOB	DateTime	AN		
Branch name	varchar (100)	NN		
Contact	number (10,0)	AN		

Table 4.13 Table: Camp

Column	Data Type	Null	Keys & Constrains	Default Value & Description
Camp Id	int	NN	PK (Auto Increment)	
Camp Name	varchar (100)	NN		
location	Varchar (100)	AN		
Organized date	datetime	AN		

5 Stories and Scenario

5.1 Story-1: Blood Donor Registration

Story # S1	:	As a Blood Donor, I want to register in the system, so that I can donate blood and help patients in need
Priority	:	High
Estimate	:	XL
Reason	:	Donor registration is essential to maintain a database of available blood donors, making it easier for hospitals and blood banks to contact donors when blood is required.

5.1.1 Scenario# S1.1

Scenario# S1.1	:	Donor Registration with Valid Information
Prerequisite	:	The donor is on the Blood Bank Management System registration page.
Acceptance Criteria	:	Given : The donor is on the donor registration page and all required fields such as name, age, blood group, contact number, and address are available. When: • The donor enters valid information in all required fields • The donor clicks the "Register" button Then : The system successfully registers the donor and displays a confirmation message indicating successful registration.

5.1.2 Scenario# S1.2

Scenario# S1.2	:	Donor Registration with Invalid Information.
Prerequisite	:	The donor is on the Blood Bank Management System registration page.
Acceptance Criteria	:	Given : The donor is on the registration page. When: • The donor enters incomplete or incorrect information • The donor clicks the "Register" button Then : The system displays error messages indicating missing or incorrect information and the donor registration is not completed.

5.1.3 Scenario# S1.3

Scenario# S1.3	:	Attempting Duplicate Donor Registration
Prerequisite	:	The donor is on the registration page of the Blood Bank Management System.
Acceptance Criteria	:	Given : The donor information such as phone number or email already exists in the system. When: • The donor tries to register again using the same contact details • The donor clicks the "Register" button Then : The system displays a message that the donor is already registered and prevents duplicate registration.

5.2 Story-2: Search Blood

Story # S2	:	As an Acceptor, I want to search for Blood-by-Blood Group, Donor Name So that I can quickly find Blood that match my Blood Group.
Priority	:	High
Estimate	:	M
Reason	:	Implementing a search functionality is essential for enhancing the user experience, as it allows Administrator and User to efficiently discover and access the Blood Bank's resources.

5.2.1 Scenario# S1.1

Scenario# S1.1	:	Donor Registration with Valid Information
Prerequisite	:	The donor is on the Blood Bank Management System registration page.
Acceptance Criteria	:	Given : The donor is on the donor registration page and all required fields such as name, age, blood group, contact number, and address are available. When: • The donor enters valid information in all required fields • The donor clicks the "Register" button Then : The system successfully registers the donor and displays a confirmation message indicating successful registration.

5.2.2 Scenario# S1.2

Scenario# S1.2	: Donor Registration with Invalid Information.
Prerequisite	: The donor is on the Blood Bank Management System registration page.
Acceptance Criteria	Given : The donor is on the registration page. When: • The donor enters incomplete or incorrect information • The donor clicks the "Register" button Then : The system displays error messages indicating missing or incorrect information and the donor registration is not completed.

5.2.3 Scenario# S1.3

Scenario# S1.3	: Attempting Duplicate Donor Registration
Prerequisite	: The donor is on the registration page of the Blood Bank Management System.
Acceptance Criteria	Given : The donor information such as phone number or email already exists in the system. When: • The donor tries to register again using the same contact details • The donor clicks the "Register" button Then : The system displays a message that the donor is already registered and prevents duplicate registration.

5.3 Story-3: Blood Donation Camp Management

Story # S3	: As an Admin, I want to create and manage blood donation camps, so that more donors can participate
Priority	: medium
Estimate	: L
Reason	: Donation camps increase blood collection and donor engagement.

5.3.1 Scenario# S3.1

Scenario# S3.1	:	Creating a Blood Donation Camp
Prerequisite	:	The admin is logged into the Blood Bank Management System.
Acceptance Criteria	:	Given : The admin is on the Blood Donation Camp Management page When : The admin enters camp details (camp name, date, location, organizer) The admin clicks the "Create Camp" button Then : The system successfully creates the camp The camp is displayed in the camp list A confirmation message is shown to the admin

5.3.2 Scenario# S3.2

Scenario# S3.2	:	Donor Registration for Blood Donation Camp
Prerequisite	:	The blood donation camp is already created and visible in the system.
Acceptance Criteria	:	Given : The donor is logged into the system and viewing available camps When : The donor selects a camp The donor clicks the "Register" button Then : The system registers the donor for the selected camp The donor receives confirmation (message/notification) The donor list for the camp is updated

5.3.3 Scenario# S3.3

Scenario# S3.3	:	Updating Blood Stock After Camp
Prerequisite	:	The blood donation camp has been completed.
Acceptance Criteria	:	Given : The admin is on the completed camp details page When : The admin enters collected blood units for each blood group The admin clicks the "Update Stock" button

	Then The system updates the blood bank inventory Stock levels are increased accordingly A success message is displayed	:
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5.4 Story-4: Blood Stock Management

Story # S3	:	As an Admin, I want to manage blood stock, so that I can keep track of available and used blood units
Priority	:	HIGH
Estimate	:	XL
Reason	:	Proper inventory management is necessary to avoid shortages or wastage of blood units.

5.5 Story-5: Donor Eligibility Check

Story # S3	:	As a System, I want to check donor eligibility, so that only eligible donors can donate blood
Priority	:	High
Estimate	:	M
Reason	:	Eligibility checks ensure safety and compliance with medical guidelines.

6 Test cases

Project Name:	Blood Bank management system	Test Designed by:	T.P.Bavaliya
Module Name:	Blood Request	Test Designed date:	16-2-2026
Release Version:	1.0	Test Executed by:	R. B. Gondaliya
		Test Execution date:	15-08-2026

Pre-condition: Web application should be accessible

Test Case ID	Test Title	Test Type	Description	Test Case ID
TC_001	Login to web application with valid credential	Functional	Login to Blood Bank management system web application through valid credential	TC_001
TC_002	Login to web application with invalid credential	Functional	Login to Blood Bank management system web application through invalid credential	TC_002
TC_003	Verify login page elements	GUI	verify that all elements are available on login page	TC_003

Test Case Title	Create Blood Request with Valid Details
Test Type	Functional
Test Priority	High
Pre-condition	User (patient/admin) should be logged into the system

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	BUG ID
1	Navigate to Blood Request Page	Blood request page should open successfully	Page opened successfully	Pass		Dashboard → Request Blood	
2	Enter Patient Name	Field should accept valid patient name	Input accepted	Pass		Patient Name: Rahul Patel	
3	Select Blood Group	Dropdown should allow selection	Blood group selected	pass		Blood Group: B+	
4	Enter Required Units	Field should accept numeric values only	Input accepted	Pass		Units: 2	
5	Enter Contact Details	Field should accept valid phone/email	Input accepted	pass		Phone: 9999999210	

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6	Enter Hospital Name	Field should accept hospital name	Input accepted	pass		Hospital: Darshan Hospital	
7	Click Submit button	Request should be submitted successfully	Request submitted and confirmation shown	pass			
8	Verify Request in List	New request should appear in request list	Request visible in list	pass			

Test Case Title	Create Blood Request with inValid Details
Test Type	Functional
Test Priority	Medium
Pre-condition	User (patient/admin) should be logged into the system

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	Bug ID
1	Leave Patient Name empty	System should validate input	Validation error shown	Pass	Required field validation		
2	Do not select Blood Group	System should prevent submission	Error message displayed	Pass			
3	Leave Units field empty	System should show error message	Field accepted empty value	Fail	Required field not enforced		BUG-101
4		Request should not be submitted	Submission blocked	Pass	"All fields are required" message		

Test Case Title	Verify Search Blood Page Elements and Functionality
Test Type	GUI

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Test Priority	High
Pre-condition	Blood Bank Management System should be accessible and the user should be on the Search/Home page.

test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	Bug ID
1	Launch website with the given URL	The site should launch properly	Site launched successfully	Pass			
2	Verify presence of search elements (Blood Group dropdown, City/Area field, Search button)	All UI controls should be visible and aligned	All elements displayed correctly	Pass			
3	Verify that the Blood Group dropdown contains all groups (A+, A-, B+, B-, O+, O-, AB+, AB-)	Dropdown should list all 8 blood groups	All options are available	Pass		Dropdown Menu	
4	Verify that the "Search" button is enabled by default	Button should be clickable	Button is clickable	Pass			
5	Enter a valid Blood Group and City, then click Search	System should display a list of available donors or bags	Results displayed in a table/list	Pass		O+, New York	
6	Verify that the results show Donor Name, Contact, and Availability Status	All necessary data columns should be visible	Data columns are visible	Pass			

Test Case Title	Verify Search Blood invalid Page Elements and Functionality
Test Type	GUI
Test Priority	High
Pre-condition	Blood Bank Management System should be accessible and the user should be on the Search/Home page.

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	Bug ID
1	Click Search without entering any data	System should show validation message	No validation message displayed	Fail	Validation missing		BUG-701
2	Enter only Blood Group without Location	System should require all mandatory fields	Search executed successfully	Fail	Incomplete validation	Blood Group: A+	BUG-702
3	Enter invalid characters in Location field	System should restrict invalid input	Special characters accepted	Fail	Input validation issue	@@@@	BUG-703
4	Enter negative or invalid value (if units/filter exists)	System should validate input	Invalid value accepted	Fail	Validation missing	-5	BUG-704
5	Enter extremely long text in Location field	System should limit input or show error	System accepts long input	Fail	No length validation	200+ characters	BUG-705
6	Search for unavailable blood group	System should show "No Data Found" message	Blank page displayed	Fail	Message missing	AB-	BUG-706

Test Case Title	Make Payment with Valid Details
Test Type	Functional
Test Priority	High

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Pre-condition	User should be logged in and on Payment page
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Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	Bug ID
1	Navigate to Payment page	Payment page should load successfully	Page loaded successfully	Pass		/payment	
2	Enter valid Card Number	Field should accept valid card number	Card number accepted	Pass		4111 1111 1111 1111	
3	Enter valid Expiry Date	Field should accept valid date	Expiry accepted	Pass		12/28	
4	Enter valid CVV	CVV should be masked and accept input	CVV accepted	Pass		123	
5	Enter Card Holder Name	Field should accept valid name	Name accepted	Pass		Rahul Patel	
6	Click Pay Now button	Payment should be successful	Payment completed successfully	Pass			
7	Verify confirmation message	Success message should display	Payment success message shown	Pass			

Test Case Title	Payment with Invalid Card Number
Test Type	Functional
Test Priority	High

Test Step	Test Case Description	Expected Result	Actual Result	Status	Comment	Data	Bug ID
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1	Enter invalid Card Number	System should show validation error	Invalid card accepted	Fail	Card validation missing	1234 5678	BUG-801
2	Enter valid other details	Fields should accept input	Input accepted	Pass			
3	Click Pay Now	Payment should not proceed	Payment processed	Fail	Critical issue		BUG-801

7 References

- <https://mybbms.netlify.app/>
- https://akshaynagpal.github.io/BloodBankManagementSystem_Version_1.1/
- <https://www.phpscriptsonline.com/product/online-blood-donor-website>
- <https://www.bloodcentresoftware.com/>

BBMS_Tushar

Product concept

Access current weather data for any location on Earth! We collect and process weather data from different sources such as global and local weather models, satellites, radars and a vast network of weather stations. Data is available in JSON, XML, or HTML format.

GET GET

`https://api.openweathermap.org/data/2.5/weather?lat=22.303894&lon=70.8&appid=766a5841767bd542fd9f27e65a266547`

PARAMS

lat	22.303894
	Latitude. If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
lon	70.8
	Longitude. If you need the geocoder to automatic convert city names and zip-codes to geo coordinates and the other way around, please use our Geocoding API
appid	766a5841767bd542fd9f27e65a266547
	Your unique API key (you can always find it on your account page under the "API key" tab)

POST POST

`https://api.openweathermap.org/data/2.5/weather?lat=22.303894&lon=70.8&appid=766a5841767bd542fd9f27e65a266547`

PARAMS

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appid**766a5841767bd542fd9f27e65a266547**

Your unique API key (you can always find it on your account page under the "API key" tab)

PUT PUT

<https://api.openweathermap.org/data/2.5/weather?lat=22.303894&lon=70.8&appid=766a5841767bd542fd9f27e65a266547>

PARAMS

lat**22.303894**

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appid**766a5841767bd542fd9f27e65a266547**

Your unique API key (you can always find it on your account page under the "API key" tab)

DELETE DELETE

<https://api.openweathermap.org/data/2.5/weather?lat=22.303894&lon=70.8&appid=766a5841767bd542fd9f27e65a266547>

PARAMS

lat**22.303894**

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